If you look at the loss plot for a *generative adversarial network* (GAN), the generator and discriminator loss look somewhat similar to our DQN and forward model loss with `use_extrinsic=False`. The losses do not smoothly decrease like you're used to when you train a single machine learning model.

A better assessment of how well the overall training is going is to track the episode length over time. The episode length should be increasing if the agent is learning how to progress through the environment more effectively. In our training loop, whenever the episode finishes (i.e., when the `done` variable becomes `True` because the agent dies or doesn't make sufficient forward progress), we save the current `info['x_pos']` to the `ep_lengths` list. We expect that the maximum episode lengths will get longer and longer over training time.

```
>>> plt.figure()
>>> plt.plot(np.array(ep_lengths), label='Episode length')
```

In figure 8.19 we see that early on the biggest spike is getting to the 150 mark (i.e., the x position in the game), but over training time the farthest distance the agent is able to reach (represented by the height of the spikes) steadily increases, although there is some randomness.

Figure 8.19. Training time is on the x axis and episode length is on the y axis. We see bigger and bigger spikes over training time, which is what we expect.